# DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW

**Evaluation Scheme & Syllabus**
**For**
**B.Tech. 2nd Year**

# MINING ENGINEERING

(Effective from session 2024-25)

## DR. A. P. J. ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW

**SEMESTER –III**

| SN | Subject Code | Subject | Type | Category | Periods | | | Sessional Component | | Sessional) SW (TS/PS | End Semester Examination ESE ( | Total SW+ESE | Credit Cr |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | L | T | P | CT | TA | CT+TA | TE/PE | | |
| 1 | BOE3**/ BAS302 | Science Based Open Elective/ BSC(Maths-III/Math IV/ Math V) | T | ES/BS | 3 | 1 | 0 | 20 | 10 | 30 | 70 | 100 | 4 |
| 2 | BVE301 / BAS301 | Universal Human Value and Professional Ethics / Technical Communication | T | VA/HS | 2 | 1 | 0 | 20 | 10 | 30 | 70 | 100 | 3 |
| 3 | BMN301 | Mineralogy, Petrology And Stratigraphy | T | PC | 3 | 1 | 0 | 20 | 10 | 30 | 70 | 100 | 4 |
| 4 | BMN302 | Mechanical Engineering | T | PC | 3 | 1 | 0 | 20 | 10 | 30 | 70 | 100 | 4 |
| 5 | BMN303 | Mine Surveying – I | T | PC | 2 | 1 | 0 | 20 | 10 | 30 | 70 | 100 | 3 |
| 6 | BMN351 | Mineralogy And Petrology Lab | P | PC | 0 | 0 | 2 | | 50 | 50 | 50 | 100 | 1 |
| 7 | BMN352 | Mechanical Engineering Lab | P | PC | 0 | 0 | 2 | | 50 | 50 | 50 | 100 | 1 |
| 8 | BMN353 | Mine Surveying-I Lab | P | PC | 0 | 0 | 2 | | 50 | 50 | 50 | 100 | 1 |
| 10 | BCC301 / BCC302 | Cyber Security/ Python Programming | T | VA | 2 | 0 | 0 | 20 | 10 | 30 | 70 | 100 | 2 |
| 11 | BCC351 | Internship Assessment /Mini Project* | P | | | | | | | 100 | | 100 | 2 |
| | | **Total** | | | **15** | **5** | **6** | | | | | | **25** |

- **Mathematics –III** for CE / ENV/ Mining and allied branches
- **Mathematics-IV** for Computer/Electronics/Electrical & allied Branches, Mechanical & Allied Branches Textile/Chemical & allied Branches
- **Mathematics-V** for Bio Technology / Agriculture Engineering

## SEMESTER –IV

| SN | Subject Code | Subject | Type | Category | Periods | | | Sessional Component | | Sessional)SW TS/PS ( | End Semester Examination ESE | Total SW+ESE | Credit Cr |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | L | T | P | CT | TA | CT+TA | TE/PE | | |
| 1 | BAS402 / BOE4** | BSC(Maths-III/MathIV/Math V)/Science Based Open Elective | T | BS/ES | 3 | 1 | 0 | 20 | 10 | 30 | 70 | 100 | 4 |
| 2 | BAS401 /BVE401 | Technical Communication / Universal Human Value and Professional Ethics | T | HS/VA | 2 | 1 | 0 | 20 | 10 | 30 | 70 | 100 | 3 |
| 3 | BMN401 | Mine Surveying – II | T | PC | 3 | 1 | 0 | 20 | 10 | 30 | 70 | 100 | 4 |
| 4 | BMN402 | Geology For Mining Engineers | T | PC | 3 | 1 | 0 | 20 | 10 | 30 | 70 | 100 | 4 |
| 5 | BMN403 | Elements of Mining | T | PC | 2 | 1 | 0 | 20 | 10 | 30 | 70 | 100 | 3 |
| 6 | BMN451 | Mine Surveying – II Lab | P | PC | 0 | 0 | 2 | | 50 | 50 | 50 | 100 | 1 |
| 7 | BMN452 | Geology For Mining Engineers Lab | P | PC | 0 | 0 | 2 | | 50 | 50 | 50 | 100 | 1 |
| 8 | BMN453 | Elements of Mining Lab | P | PC | 0 | 0 | 2 | | 50 | 50 | 50 | 100 | 1 |
| 9 | BCC401/ BCC402 | Cyber Security/ Python Programming | P | VA | 2 | 0 | 0 | 20 | 10 | 30 | 70 | 100 | 2 |
| 10 | BVE451 / BVE452 | Sports and Yoga - II / NSS-II | P | VA | 0 | 0 | 3 | | | 100 | | 100 | 0 |
| | | **Total** | | | **15** | **5** | **9** | | | | | | **23** |
| | | **Minor Degree/ Honors Degree MT-1/HT-1** | | | | | | | | | | | |

***The Mini Project or internship (4 weeks) will be done during summer break after 4th Semester and will be assessed during V semester.**

## BMN301- MINERALOGY, PETROLOGY AND STRATIGRAPHY

**Course Objectives:**

1. Understand physical and chemical properties of minerals, including their classification.
2. Explain the origin and characteristics of igneous, sedimentary, and metamorphic rocks.
3. Gain knowledge of paleontology, fossil preservation, and their role in interpreting Earth's history.
4. Learn principles of stratigraphy and their application in identifying important geological resources in India.
5. Study the processes shaping the Earth's crust and the distribution of natural resources.

**Course Outcomes:**

The student will able to:

1. Classify common silicate, sulphide, and oxide minerals.
2. Differentiate between rock types based on their formation and textures.
3. Understand sedimentation, metamorphism, and their significance in the rock cycle.
4. Interpret fossil records to analyze geologic time and environmental changes.
5. Apply stratigraphic principles to recognize rock formations and resources in different settings, especially in India.
6. Build a foundation for advanced studies in geology and resource exploration.

**UNIT I (08 Lecture)**

**Minerals:** Physical and chemical properties; Crystal, crystal classes and systems; Classification of minerals and properties of common silicate minerals (Quartz, Feldspar, Pyroxene, Amphibole, Garnet, Olivine, Mica), sulphides (Pyrite, Chalcopyrite, Galena, Sphalerite) and oxides (Haematite, Magnetite, Chromite, Pyrolusite, Psilomelane). **UNIT II (08 Lecture)**

**Igneous rocks:** Magma and lava, extrusive and intrusive forms, textures; Classification and description of some common igneous rocks (Granite, Dolerite, gabbro, Basalt, Rhyolite, Pegmatite). **UNIT III (08 Lecture)**

**Sedimentary rocks**: Sedimentation processes; Classification and description of some common sedimentary rocks (Conglomerate, Sandstone, Shale, Limestone). **UNIT IV (08 Lecture)**

**Metamorphic rocks:** Processes of metamorphism, textures and structures of metamorphic rocks; Classification and description of some common metamorphic rocks (Slate, Phyllite, Schist, Gneiss, Quartzite, Marble). **UNIT V (08 Lecture)**

**Paleontology and Stratigraphy:** Concepts of palaeontology; Fossils, their mode of preservation and significance as indices of age and climate; Concept of index fossils. Principles of stratigraphy; Broad stratigraphic subdivisions and associated rock types of important ore provinces, coal belts and oil fields of India.

**Books and References:**


1. Parbin Singh-A Text Book of Engineering & General Geology- S.K.Kataria & sons
2. N Chenna Kesavulu- A Text book of Engineering Geology- Macmillan India Ltd.
3. M.T.Maruthesha Reddy- A Text book of Applied Engineering Geology- New Age International Publisher
4. Mukerjee P.K. , A Textbook of Geology, The World Press Pvt.LtD Calcutta.
5. Tyrell G.W.,The Principles of Petrology, B.I.Publications Pvt. Ltd

6. H.H.Read ,Textbook of Minerology Oxford University Press.Delhi

## BMN 302- MECHANICAL ENGINEERING

**Course Objectives:**

1. Understand stress-strain behavior, elastic constants, and principal stresses.
2. Learn beam deflection, pressure vessel stresses, and torsion analysis.
3. Grasp basic machine theory, gears, mechanisms, and machine elements.
4. Analyze thermodynamic processes and air standard cycles.
5. Study gas turbines, air compressors, and internal combustion engine performance.

**Course Outcomes:**

The student will able to:

1. Interpret stress-strain diagrams and calculate thermal stresses.
2. Analyze beam deflection, pressure vessels, and shaft torsion.
3. Evaluate mechanisms, gear trains, and machine element performance.
4. Apply thermodynamic principles to air standard cycles and processes.
5. Assess the performance of engines, compressors, and turbines.
6. Conduct case studies based on lab setups for real-world applications.

**UNIT I (08 Lecture)**

**Introduction to Strength of Materials:** Stress-strain diagram; Elastic constants and their relations, Thermal stresses and strains, Stresses in oblique planes - Principal stresses and principal planes.

**UNIT II (08 Lecture)**

**Theory of simple bending:** Deflection of beams-integration method and moment area method. Analysis of stresses in pressure vessels- thin and thick cylinders. Torsion of solid and hollow circular Shafts.

**UNIT III (08 Lecture)**

**Introduction to theory of Machines:** Basic concepts: degrees of freedom, kinematic constrains, linkages, mechanisms. Different types of gears, gear trains, reduction ratio and torque assessment, application of gearboxes. Basic principles and constructions of governors, flywheels, brakes, clutches and dynamometers.

**UNIT IV (08 Lecture)**

**Introduction to thermodynamics;** Analysis of various thermodynamic processes, P-V and T-S diagrams. Analysis of air standard cycles – Otto, Diesel and Dual cycles. Classifications, applications and performance estimation of internal combustion engines; **UNIT V (08 Lecture)**

Fundamentals of simple open cycle and closed cycle gas turbines and reciprocating air compressors – single and multi-stage. Performance study and power estimation on laboratory experimental data. Case study based on laboratory setups.

**Books and References:**

1. Mechanics of Materials – Beer, Johnston et-al.
2. Strength of Materials – S.S. Rattan.
3. Strength of Materials – V.S. Prasad.
4. Theory of Machines – Sadhu Singh.

5. Theory of Machines – S.S. Rattan.
6. Theory of Machines and Mechanisms – Uicker, Pennock and Sigley.
7. Engg. Thermodynamics – P.K.Nag.
8. Thermodynamics – Cengel and Boles.
9. Thermal Engg. - Rathore

## BMN303- MINE SURVEYING – I

**Course Objectives:**

1. Understand the basics, objectives, and principles of surveying.
2. Learn distance and angular measurement techniques using surveying instruments.
3. Master levelling methods and use of levelling instruments.
4. Study Total Station principles and applications in surveying.
5. Grasp contouring concepts and perform plane table surveying for area and volume computation.

**Course Outcomes:**
The student will able to:

1. Understand surveying principles and perform accurate linear measurements.
2. Apply angular measurement techniques and adjust traverses.
3. Perform various levelling operations and data reduction.
4. Use Total Station for precise distance and angle measurements.
5. Conduct contouring and plane table surveys for area and volume calculations.

**UNIT I (8 Hours)**

**Surveying:** Definition, objective, classification and principles of surveying.

**Linear Measurement:** Instruments for measuring distances; Ranging and taping survey lines; Chain surveying – principle, field work, off-sets, booking and plotting, obstacles in taping. **UNIT II (8 Hours)**

**Angular Measurement:** Bearing of lines; Rectangular coordinate system; Essentials of the microoptic theodolite; Measurement of horizontal and vertical angles; Temporary and permanent adjustments; Theodolite traversing; Computation of co- ordinates; Adjustment of traverse; Temporary and permanent adjustments. **UNIT III (8 Hours)**

**Levelling:** Definition & terminology; Levelling instruments types - tilting, auto set and digital levels; Levelling staves; Different types of levelling - differential, profile, cross-sectional and reciprocal levelling; Booking and reduction methods; Underground levelling; Temporary and permanent adjustments of levels.

**UNIT IV (8 Hours)**

**Total Station:** Principle of electronic measurement of distance and angles; construction and working with Total Station; Errors; Application and recent developments in Total Station. **UNIT V (8 Hours)**

**Contours: Concepts;** Characteristics of contour; Contour Interval; Methods of contouring and uses of contours.

**Plane Table Surveying:** Methods; Detail surveying and contouring using plane table and microoptic alidade, Computation of areas and volumes.

Text Books

1. Punmia, B. C. (2005), Surveying Vol. 1 and II
2. Schofield, W. and Breach M. (2006), Engineering Surveying

**Reference Books**

1. Bannister, A. and Raymond S., Surveying
2. Advances in Surveying Technology: Lecture Notes by faculty

## BMN351-MINERALOGY AND PETROLOGY LAB

**List of Experiments:**

Study of physical properties of:

1. Rock forming minerals: Talc, Gypsum, Calcite, Fluorite, Feldspar (Orthoclase, Microcline, Plagioclase), Muscovite, Biotite, Quartz, Beryl, Tourmaline, 2. Corundum, Kyanite, Serpentine, Garnet and Sillimani.
3. Ore minerals: Haematite, Magnetite. Chalcopyrite, Malachite,
4. Azurite, Chromite, Bauxite, Pyrolusite, Psilomelane, Sphalerite, Galena

**Study of common rocks with reference to their structures, mineral composition and uses.**

5. Igneous Rocks: Granite, Syenite, Gabbro, Basalt, Dolerite, Lamprophyre, Aplite, Pegmatite.
6. Metamorphic Rocks: Slate, Schists, Gneisses, Quartzite, Marble, Amphibolite, Charnockite.
7. Sedimentary Rocks: Conglomerate, Sandstone, Shale.
8. Carbonaceous Shale, Coal, Limestone.

## BMN 352-MECHANICAL ENGINEERING – I LAB

**List of Experiments:**

1. Tensile test of a mild steel specimen on Universal Testing Machine (UTM)
2. Young's modulus by flexure of beam method
3. Stiffness test of closed coiled helical spring
4. Determination of M. I. of a connecting rod
5. Coefficient of friction between belt / rope and pulley
6. Efficiency of a screw jack
7. Study of construction and operation of 4-stroke SI engine model.
8. Study of construction and operation of 4-stroke CI engine model.
9. Performance testing of a 4-stroke Diesel engine.

## BMN-353 MINE SURVEYING – I LAB

1. To prepare conventional symbol chart based on the study of different types of topographical maps.
2. To measure bearings of a closed traverse by prismatic compass and to adjust the traverse by graphical method.
3. To find out reduced levels of given points using Auto/dumpy level.
4. To perform fly leveling with Auto/tilting level.
5. To study parts of a Vernier theodolite and measurement of horizontal and vertical angle.
6. To measure horizontal angle between two objects by repetition/reiteration method.
7. To determine the height of a vertical structure (e.g. chimney/ water tank etc.) using trigonometrical levelling by taking observations in single vertical plane.

8. To study various parts of Electronic Theodolite, Total Station and practice for measurement of distance, horizontal and vertical angles.
9. To set out a simple circular curve by Rankine's method

## BMN401 MINE SURVEYING II Course

**Objectives:**

1. Learn control survey techniques, triangulation, and astronomical direction measurement.
2. Understand Gyro-theodolite principles and correlation methods in shaft surveys.
3. Study development and stope surveys in mining for direction and gradient control.
4. Explore slope monitoring and GPS applications in mine surveying.
5. Familiarize with automation, surveying software, GIS, and remote sensing in surveying.

**Course Outcomes:**

The student will able to:

1. Conduct triangulation and direction measurement using astronomical observations.
2. Apply Gyro-theodolite methods for shaft surveying and correlation.
3. Perform development and stope surveys in mining environments.
4. Monitor slopes in mines using GPS, remote sensing, and stability techniques.
5. Use surveying software, GIS, and automation for data collection and earthwork calculation.

**UNIT I (8 Hours)**

**Control Surveys:** Triangulation – classification; Reconnaissance; Procedures for angles and baseline measurement; Comparison with precise EDM traversing.

**Direction Measurement by Astronomical observation:** Definition of astronomical terms; Time systems; Determination of true bearing of a survey line by astronomical observations. **UNIT II (8 Hours)**

**Gyro-North Determination:** Principle of Gyro-theodolite/Gyromat; Determination of Gyro-north.

**Correlation:** Methods of correlation – direct traversing in inclined shaft, correlation in vertical shaft – single and two shafts, Gyro-Laser combination; Shaft depth measurement. **UNIT III (8 Hours)**

**Development Surveys:** Control of direction and gradient in drifts, tunnels, raises, winzes. **Stope Surveying:** Purpose; Methods of survey in moderately and steeply inclined ore bodies, flat and vertical ore bodies/seams. **UNIT IV (8 Hours)**

**Slope Monitoring in Opencast Mines**: Geodetic and Remote Sensing Methods, Slope Stability Radars

**GPS**: Principle of GPS; Instrument; Errors and working with GPS; Application of GPS in mine surveying; Developments in satellite based Navigation system. **UNIT V (8 Hours)**

**Application of Automation & IT in surveying:** Data acquisitions; Preparation of plans and sections; Calculation of earth works

**Introduction to Surveying softwares**

Application of GIS and Remote Sensing in Surveying. **Text Books**

1. Punmia, B. C. (2005), Surveying Vol. 1 and II
2. Schofield, W. and Breach M. (2006), Engineering Surveying

**Reference Books**

1. Bannister, A. and Raymond S., Surveying
2. Advances in Surveying Technology: Lecture Notes by faculty

## BMN402 Geology for Mining Engineers

**Course Objectives:**

1. Study topographic maps and geological structures like folds and faults.
2. Learn about igneous intrusions, their effects on strata, and stereographic projections.
3. Understand economic geology, ore formation, and major Indian mineral deposits.
4. Explore mineral exploration methods, strategies, and stages.
5. Examine coal and petroleum geology, including coal characteristics and petroleum basins in India.

**Course Outcomes:**

The student will able to:

1. Interpret topographic maps and recognize geological structures.
2. Identify igneous intrusions and use stereographic projections.
3. Understand ore formation processes and Indian mineral deposits.
4. Apply mineral exploration techniques and strategies.
5. Analyze coal properties and petroleum geology in India.

**UNIT I (8 Hours)**

**Structural Geology:** Study of topographic maps; Attitude of planar and linear structures; Effects of topography on outcrops. Unconformities, folds, faults and joints - their nomenclature, classification and recognition.

**UNIT II (8 Hours)**

**Forms of igneous intrusions** - dyke, sill and batholith. Effects of folds and fractures on strata/orebodies and their importance in mining operations. Principles of stereographic projections of linear and planar features of rocks. **UNIT III (8 Hours)**

**Economic Geology and Exploration Geology:** Introduction and scope of economic geology; Ore and gangue; Processes of ore formation; Major Indian mineral deposits (Iron, Manganese, Copper, Lead, Zinc) - distribution and mode of occurrence. **UNIT IV (8 Hours)**

**Mineral Exploration** – concepts and methods viz. surface and subsurface; Exploration strategy and design; Stages of exploration; Resources and reserves. UNIT V (8 Hours)

**Coal and Petroleum Geology:** Rank, characteristics and important constituents of coal; Classification and origin of coal; Chief characteristics of Indian coals; Geology of the principal coalfields of India. Concept of organic constituents of petroleum origin, migration, accumulation, concept of traps and important petroliferous basins of India.

**Books and References:**

1. Peters, W. C. Exploration and Mining Geology (2nd Ed.); 1987. John Wiley & Sons, New York.
2. P.K. Mukerjee, A Text Book of Geology (13th Ed.); 1997. Reprinted 2018, The World Press Pvt. Ltd. Kolkata.
3. RNP Arogyaswamy, Courses in Mining Geology (Fourth Edition) 1995, Reprint, 2017, OXFORD & IBH; CBS Publishers & Distributors Pvt. Ltd

**REFERENCE BOOKS:**

1. Geology and Mineral Resources of India, Misc. Publication No. 30, Part- XXII,
2. Geological Survey of India, 2010.

3. Mining Geology by Hugh Exton Mc Kinstry, 1948, Asia Publishing House (1960) 4. Geological Methods in Mineral Exploration and Mining by Roger Marjoribanks, Springer 2nd Ed. 2010.
5. Introduction to Mineral Exploration, Second Edition, Edited by Charles J. Moon, Michael K. G. Whateley & Anthony M. Evans

## BMN403 ELEMENTS OF MINING

**Course Objectives:**

1. Understand mining types, life cycle, exploratory drilling, and Indian mineral resources.
2. Learn methods for opening deposits, including entry modes and box cuts.
3. Study shaft construction and incline/decline drivage methods.
4. Review underground mining methods, including coal and metal mining, and basic mine infrastructure.
5. Explore surface mining methods, equipment selection, and pit geometry.

**Course Outcomes:**

The student will able to:

1. Define and classify mines, and understand their life cycle and resources.
2. Apply methods for deposit opening and box cut location.
3. Describe shaft construction and incline/decline methods.
4. Compare underground mining methods and understand mine infrastructure.
5. Analyze surface mining techniques, equipment, and pit layout.

**Unit I (8 Hours)**

**Mining :** Definition and economic importance; Mine – definition, different types and classification; Mine life cycle, Exploratory Drilling; Mineral deposit – different types and their classification; Mineral resources of India **Unit II (8 Hours)**

**Opening-up of Deposits :** Choice of mode of entry - adit, shaft, incline, decline and combined mode, their applicability, number and disposition. Box cut- types and location **Unit III (8 Hours)**

**Vertical/Inclined Shafts, Inclines/Declines :** Location, shape, size, and organisation of shaft sinking, sinking methods, construction of shaft collar, shaft inset, shaft fittings.
Methods of incline/decline drivage. **Unit**

**IV (8 Hours)**

**Overview of underground mining :** Different coal mining methods and their applicability & limitations; Different metal mining methods and their applicability & limitations; Basic concepts of transportation, ventilation, illumination and support in underground mines. **Unit V (8 Hours)**

**Overview of surface mining:** Types of surface mines, applicability & limitations, unit operations and equipment selection, pit geometry and layout.

Text Books:

1. Introductory Mining by H L Hartman
2. Elements of Mining Technology (Volume 1, 2 and 3) by D J Deshmukh 3. Mine Transport – Karelin
4. Mining and Transport – Walker.

## BMN- 451 MINE SURVEYING – II PRACTICAL

1. **Precise Levelling** and distribution of closing error.
2. Study and sketch of **Digital Level.**
3. Detailed surveying and contouring with **Total Station.**
4. Working with **SURPAC** and preparation of Mine Plan.
5. Earthwork calculation using SURPAC and plotting.
6. Plotting of sections using **LISCAD.**
7. Study and practice of **Gyro-theodolite/Gyromat.**
8. Study of **GPS** and data collection; GPS data downloading and post- processing.
9. Study of **Nadir Plummet** and LASER for Correlation survey.
10. Subsidence monitoring using 1" Micro-optic Theodolite & Precise Level.

## BMN-452 GEOLOGY FOR MINING ENGINEERS LAB

1. Minerals under handspecimen
2. Rocks under handspecimen
3. Geomorphic models
4. Crystal models under handspecimen
5. Geological succession of India and available minerals
6. Plotting of geological section along given section line in the given geologic map, Plotting Indian geological formations & mineral deposits.
7. Plotting of earthquake/ seismic belts of world and India, Plotting of volcanic belts of World and India.
8. Plotting of Physiographic maps of India, Plotting of structure/ tectonic map of India.
9. Structural models under hand specimen.
10. Interpretation of Ariel photographs with the help of stereoscope, Interpretation of the satellite imagery.

Reference Books:


1. Principles and Practices of Coal Mining by R D Singh
2. SME Mining Engineering Handbook, 3rd Edition by Peter Darling


## BMN453 Elements of Mining Lab

1. Mapping of different mineral resources of (i) Uttar Pradesh (ii) India and (iii) world.
2. Illustration of Mining Terminology
3. Various types of fishing tools used in exploratory boring for mineral and oil
4. Bore-hole logging and interpretation of bore hole data and numerical problems related to it
5. Designing of explosive magazine, Preparation of prospecting scheme.
6. Study of portable borehole machine, Study of various types of bits used in borehole coring
7. To feed the bore hole data in computer software and to take results related with the formation and quantum of mineral resources
8. Selection of various types of blasting accessories used in mines
9. PMS Plants with various capacities for surface mines